Huawei Access Node BTM-ACC-23: QoS Congestion Event SOP (ALARM_2069)

Node: BTM-ACC-23 | **Vendor:** Huawei | **Severity:** Critical | **Alarm Code:** ALARM_2069

Description

This Standard Operating Procedure (SOP) addresses a critical QoS congestion event on Huawei Access Node BTM-ACC-23, situated at BTM Layout Aggregation. The congestion event, characterized by microbursts, interface tail drops, and QoS queue starvation (ALARM_2069), is a direct consequence of inefficient Quality of Service (QoS) configuration and/or hardware limitations on the respective access node. The Huawei equipment's Role-based access control architecture facilitates the isolation of QoS policies to specific user roles, thereby enabling the prioritization of critical traffic flows. However, misconfigured or malfunctioning QoS policies can lead to an imbalance in bandwidth allocation, causing congestion and resulting in the aforementioned alarm. Furthermore, the presence of microbursts can be attributed to the interaction between the line card's buffer management and the software's policing mechanism, which can lead to tail drops and subsequent congestion. In this SOP, we will identify and rectify the root cause of the QoS congestion event, ensuring the optimal functioning of the access node and the associated network services.

Pre-requisite Checks

- Verify site accessibility and ensure safe working conditions at BTM Layout Aggregation.
- Check the status of redundant links and ensure that they are operational and available for switchover.
- Gather necessary information and documentation, including system logs and configuration files.

Potential Root Causes

Cause	Probability
Misconfigured QoS policies	High
Hardware limitations (buffer size, policing rate)	Medium
Software bugs or glitches	Low

Diagnostic Commands & Expected Outputs

```
Command: display qos policy, Expected Output: Display current QoS policies and their configurations., Description: This command will provide an overview of the QoS policies in effect on the access node, allowing us to identify any potential misconfigurations or anomalies.

Command: display interface brief, Expected Output: Display brief information about all interfaces, including their operational status and traffic statistics., Description: This command will help us determine if the congestion event is interface-specific or if it is a broader network issue.
```

```
Command: display buffer management, Expected Output: Display buffer management configuration and statistics., Description: This command will provide insights into the buffer management settings and statistics, enabling us to determine if the microbursts are caused by hardware limitations or software issues.
```

Resolution Steps

Step 1: Step: Verify and adjust QoS policies, Command: undo qos policy , Description: If misconfigured QoS policies are identified, this command will undo the existing policy and allow us to reconfigure it correctly.

Step 2: Step: Adjust buffer management settings, Command: buffer management , Description: If hardware limitations are determined to be the root cause, this command will adjust the buffer size to optimize performance and prevent microbursts.

Step 3: Step: Upgrade or patch software, Command: upgrade software package , Description: If software bugs or glitches are identified, this command will upgrade or patch the software to the latest version, resolving the issue.

Post-Check Verification

```
Command: display qos policy, Expected Output: Display current QoS policies and their configurations., Description: This command will verify that the QoS policies have been correctly reconfigured.

Command: display interface brief, Expected Output: Display brief information about all interfaces, including their operational status and traffic statistics., Description: This command will confirm that the congestion event has been resolved and that all interfaces are operating within normal parameters.

Command: display buffer management, Expected Output: Display buffer management configuration and statistics., Description: This command will verify that the buffer management settings have been correctly adjusted.
```

Roll-back Procedures

- Step: Revert QoS policies, Command: apply qos policy , Description: If the fix fails, this command will revert the QoS policies to their original state, ensuring that the network services remain operational.

- Step: Revert buffer management settings, Command: undo buffer management , Description: If the fix fails, this command will revert the buffer management settings to their original state.

- Step: Revert software changes, Command: rollback software package , Description: If the fix fails, this command will revert the software changes to their original state.